The role of budworm

Spruce budworm caterpillars have devastated balsam fir forests in eastern North America six times since the 18th century. Holling described this process as having two very different states: one with young, fast-growing trees and few budworms; and one with mature trees and very large numbers of budworms.

Between outbreaks of budworms, young balsam fir grow alongside spruce and white birch trees. Eventually, the fir becomes dominant. A combination of this dominance and a sequence of very dry years stimulates a huge increase in the budworm population. The mature fir is destroyed, giving the spruce and birch an opportunity to regenerate. By keeping the balsam fir in check, the budworm also maintains the spruce and birch. Without it, the fir trees would crowd out the others. So the system is unstable but at the same time resilient.